# Mathematical Models for Forecasting Photovoltaic Power Output

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*Abstract:* Several mathematical models for short-term photovoltaic power forecasting were developed, ranging from simple baseline approaches to advanced neural network–based models. Model performance strongly depends on the accuracy of meteorological forecasts; in our study, forecasts from the MeteoSource service proved most reliable for the investigated area. The most accurate model achieved a normalized mean absolute error (nMAE) of approximately 6.5 % when the photovoltaic power output is greater than zero. This model offers direct practical value for photovoltaic power plant operators, particularly in optimizing production forecasts and financial planning under current European spot price conditions.

*Key-Words: Photovoltaic power forecasting, Mathematical models, Physical models, Neural networks, Short-term prediction*

Received: May 31, 2019. Revised: May 4, 2020. Accepted: May 22, 2020. Published: May 29, 2020

## 1 Introduction

Renewable energy sources (RES) are becoming an increasingly important component of energy systems, driven primarily by efforts to reduce $CO_2$ emissions and diversify energy supplies. According to the International Energy Agency's Renewables 2025 report, global renewable power capacity is projected to increase by almost 4 600 GW between 2025 and 2030, roughly double the deployment seen in 2019–2024. Solar photovoltaic (PV) technologies are expected to account for nearly 80 % of this growth, highlighting the rapid expansion of solar capacity worldwide [1]. The growing penetration of PV systems introduces significant challenges for power system operation. PV output is inherently variable and strongly dependent on weather conditions, complicating grid integration, energy trading, and operational planning. Accurate short-term forecasting of PV power production is therefore crucial [2].

Several AI-based methods for energy forecasting have been proposed [4–11]. However, methodologies to assess the actual energy performance of existing PV installations remain limited. In Central Europe, PV deployment is accelerating alongside rising energy prices. In the Czech Republic, approximately 150,000 small PV systems (≈10 kWp) were installed on residential buildings by early 2025, with a total of around 170,000 PV installations and an aggregated nominal output of 3,900 MWp. Agrovoltaic systems, enabling dual land use, are also gaining importance due to sustainability considerations [12, 13].

PV power plants offer advantages such as relatively simple installation, low maintenance, and minimal overhead. Their main drawbacks are low installed power density and variability of output, largely dependent on weather conditions. Rare events like solar eclipses can also significantly affect production [14]. Accurate forecasting, leveraging weather predictions, allows operators to optimize financial returns in markets where electricity prices are dynamic and updated every 15 minutes.

Short-term PV power forecasting has been addressed in several studies. Wang et al. [4, 5] examined one-day-ahead predictions. Sadowska et al. [6] validated a real-world PV energy model for a 1 MWp farm in Poland, with a production error of 15–16 %. Su et al. [7] applied neural networks for ultra-short-term predictions, achieving similar accuracy. Wang et al. [15] compared convolutional neural networks (CNN) and long short-term memory (LSTM) networks, demonstrating that a hybrid CNN–LSTM approach yielded the best predictive performance.

## 2 Problem Formulation

The prediction model was created for a photovoltaic power plant located in the village of Kunžak in the Czech Republic (Fig. 1). It has a nominal output of 1,293 kWp. It consists of 5,625 photovoltaic panels. Each panel has a nominal output of 230 Wp with a permissible tolerance of ±5%. The panels had an initial efficiency of 14.1% at the time of installation (2010).

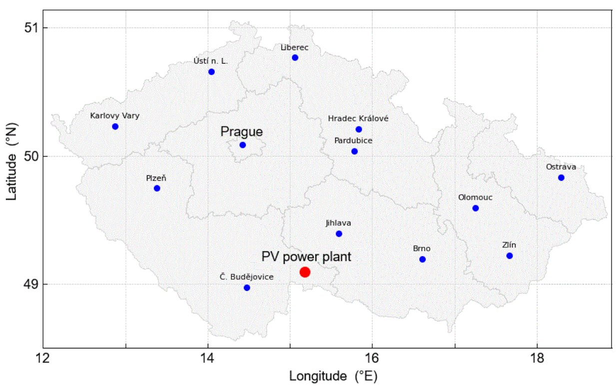


Fig. 1 Location of the PV power plant.

The dataset covers hourly measurements of aggregated PV power output and panel temperature from 2020 to 2025. It was subsequently extended with additional meteorological variables obtained from numerical weather models provided by MeteoSource. The MAE of global solar radiation for the study area is approximately 36 W/m². Further variables included atmospheric pressure, air humidity, wind speed and direction at 10 m, cloud cover, precipitation, dew point, ozone concentration, PM10, and ambient temperature. Additional features were computed using the pvlib library, including the sun's zenith, sun's azimuth and solar elevation. Temporal features capturing daily and seasonal cycles (sin_hour, cos_hour, sin_day_of_year, cos_day_of_year) were included, along with wind components (wind_u, wind_v) derived from wind speed and direction to represent east-west and north-south flow. Extensive preprocessing was performed, including correlation analysis, Yeo-Johnson transformation, standardization, and Min-Max scaling. Experiments confirmed that the combination of these features contributed most to model accuracy.

Fig. 2 shows the linear correlation of the resulting features used for the model that contribute most to the output power of the PV power plant.

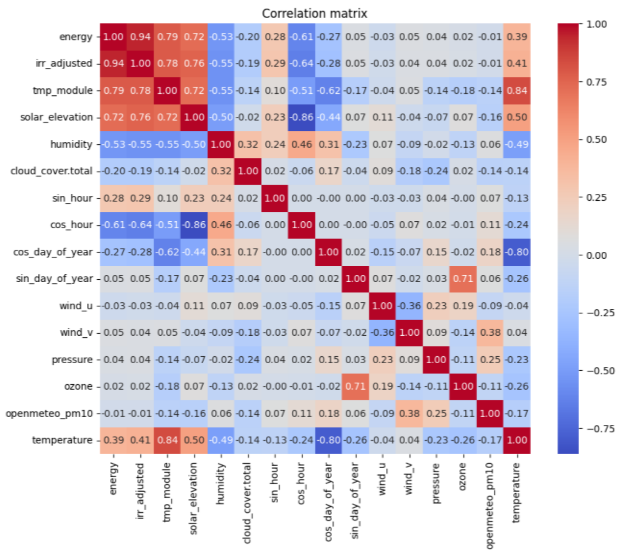


Fig. 2. Correlation matrix

Subsequently, the dataset was split into a training set (80%) and a testing set (20%), with the testing set containing 3,907 records. All performance metrics were computed only for time steps with positive power output (y_true > 0).

## 2.1 Model based on historical averages of power for each hour and day of the year

The baseline model predicts the PV output power using historical averages for each specific hour and day of the year, without considering current meteorological conditions. This approach provides a basic level of prediction. The evaluation metrics of this model are summarized in Table 1.

The model shows relatively high errors: the normalized mean absolute error (nMAE) reaches 12.48 %, the mean absolute error (MAE) is 161.41 kW, and the mean bias error (MBE) is -9.65 kW, indicating a slight systematic underestimation. The coefficient of determination ($R^2$) is 0.51, reflecting a limited ability of the model to capture the variability of the actual PV power output. Finally, the mean absolute percentage error (MAPE) reaches 159 %. These results indicate that, while the baseline model provides a rough approximation, incorporating meteorological variables is necessary to improve prediction accuracy.

Fig. 1 compares the predicted and actual PV power output for the first 200 hours of the test data, illustrating the baseline model's limited ability to capture short-term fluctuations.

Table 1. Error metrics for the historical average prediction model.

| Metrics | Value |
|---|---|
| nMAE [%] | 12.48 |
| MAE [kW] | 161.41 |
| MBE [kW] | -9.65 |
| $R^2$ [-] | 0.51 |
| MAPE [%] | 159 |

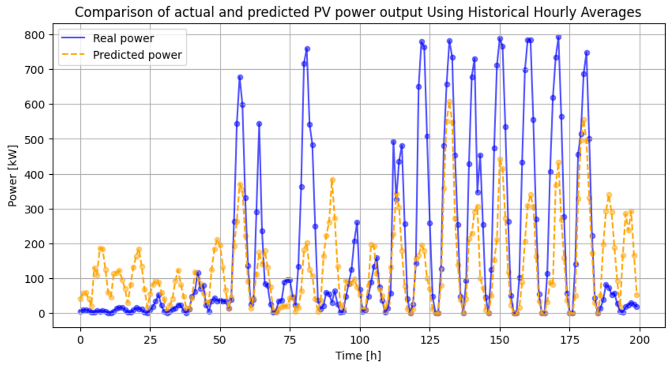


Fig. 3. Comparison of actual and predicted PV output Using Historical Hourly Averages.

## 2.2 Physical model using the basic formula for calculating the power of a PV panel

For the reference physical model, Eq. (1) was used to calculate the predicted output power of the PV power plant. Based on this equation, predicted power values for individual hours were determined. The panel inclination angle β was set to 35°, corresponding to the actual tilt of the PV panels.

$$P_{theor} = E_{total} \cdot \eta \cdot S \cdot \cos(\beta) \qquad (1)$$

$P_{theor}$ is the output power of the PV panel (W),
$E_{total}$ is the total solar radiation (W/m²) perpendicularly on the earth's surface,
η is the efficiency of the PV panel (-),
S is the area of the PV panel (m²),
β is the angle of inclination of the PV panel to the horizon (°).

The evaluation metrics of this model are summarized in Table 2. The model shows a normalized mean absolute error (nMAE) of 7.60 %, a mean absolute error (MAE) of 98.35 kW, and a mean bias error (MBE) of -59.73 kW, indicating a systematic underestimation of output power. The coefficient of determination ($R^2$) is 0.80, showing that the model captures a significant portion of the variability of actual power, while the mean absolute percentage error (MAPE) is 77.30 %. Overall, this simple physical model provides a relatively accurate approximation of PV output compared to the model based on historical averages of power for each hour and day of the year.

Fig. 4 shows the comparison between the predicted and actual PV power output for the first 200 hours of the test dataset. The model captures general trends, but significant deviations remain, reflecting its simplified assumptions.

Table 2. Error metrics for Physical model.

| Metrics | Value |
|---|---|
| nMAE [%] | 7.60 |
| MAE [kW] | 98.35 |
| MBE [kW] | -59.73 |
| $R^2$ [-] | 0.80 |
| MAPE [%] | 77.30 |

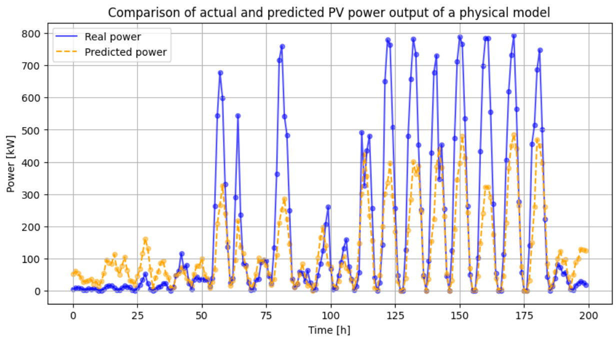


Fig. 4. Comparison of actual and Predicted PV power output of a physical model.

## 2.3 Physical model with IAM correction obtained using the pvlib library

The baseline physical model for PV power output was improved using the Incidence Angle Modifier (IAM) via the pvlib library to better account for actual solar radiation conditions. Unlike the previous model, which assumed a fixed panel tilt, the IAM correction considers the dynamic angle of incidence (AOI) between the panel surface normal and incoming solar radiation, calculated using the ASHRAE physical model along with the sun's azimuth and zenith angles. This approach captures variations in incident radiation and losses due to panel reflectivity, leading to more accurate power estimates. As shown in Table 3, the IAM-enhanced model demonstrates improved performance: the normalized mean absolute error (nMAE) is 6.6 %, MAE is 85.43 kW, MBE is -10.45 kW, $R^2$ is 0.85, and MAPE is 82.32 %. The model more closely tracks peak output, reducing systematic underestimation compared to the simpler physical model.

Fig. 5 shows the comparison between the predicted and actual PV power output for the first 200 hours of the test dataset. Although the IAM correction improves the prediction, the model still fails to fully capture the peak values, which is partly due to unmodeled system losses and panel degradation.

Table 3. Error metrics for the Physical model with IAM correction.

| Metrics | Value |
|---|---|
| nMAE [%] | 6.6 |
| MAE [kW] | 85.43 |
| MBE [kW] | -10.45 |
| $R^2$ [-] | 0.85 |
| MAPE [%] | 82.32 |

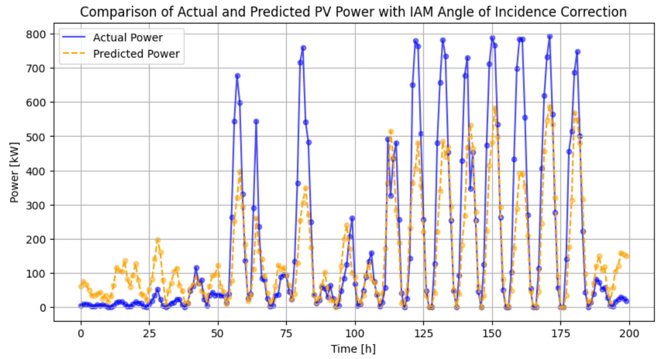


Fig. 5. Comparison of actual and predicted PV power with IAM correction.

## 2.4 Neural network-based model using only historical data

Many experiments were performed with different input features and model architectures. To ensure replicability, the same random seed was used for weight initialization in all experiments.

The most promising architecture combines a convolutional layer with an LSTM, capturing both short-term patterns and long-term dependencies. The convolutional layer extracts local correlations, while the LSTM layer models seasonal cycles.

The input tensor has dimensions of 24 time steps × 15 features, including historical PV power and meteorological measurements (global radiation, solar elevation, panel temperature, humidity, cloud cover, wind_u, wind_v, PM10, ozone, surface pressure), as well as time-related variables (sin and cos of the hour and day of the year). The architecture of this model is shown in Fig. 6.

Keras Tuner with Bayesian optimization was used for hyperparameter selection. The optimized model includes a convolutional layer with 96 filters (kernel size 2, ReLU, same padding) and an LSTM layer with 2816 units and 10 % recurrent dropout. The Dense output layer predicts 24 hours ahead with a linear activation. The model was trained using the ADAM optimizer with MAE as the loss function, a learning rate of 0.0005, and a batch size of 64. Fig. 7 shows the loss curves, where the validation curve gradually converges to the training curve, indicating that the model does not overfit. The fluctuations in the validation curve result from the use of a expanding window learning strategy.

The model achieves high accuracy, as summarized in Table 4. Normalized mean absolute error (nMAE) ranges from 7.11 % to 13.87 %, MAE from 92 kW to 179 kW, and the mean bias error (MBE) remains low, indicating minimal systematic deviation. The coefficient of determination $R^2$ decreases with forecast horizon but remains above 0.3, showing that the model captures a substantial portion of PV power variability. Compared to previous models, the neural network demonstrates substantially improved prediction performance, especially in capturing dynamic changes in output power.

Fig. 8 shows the predicted and actual PV power output for the first hour of the forecast horizon, while Fig. 9 shows the prediction for the last hour (24th) of the horizon. It is noticeable that the prediction error increases with the forecast horizon, as the model lacks information about future meteorological conditions.

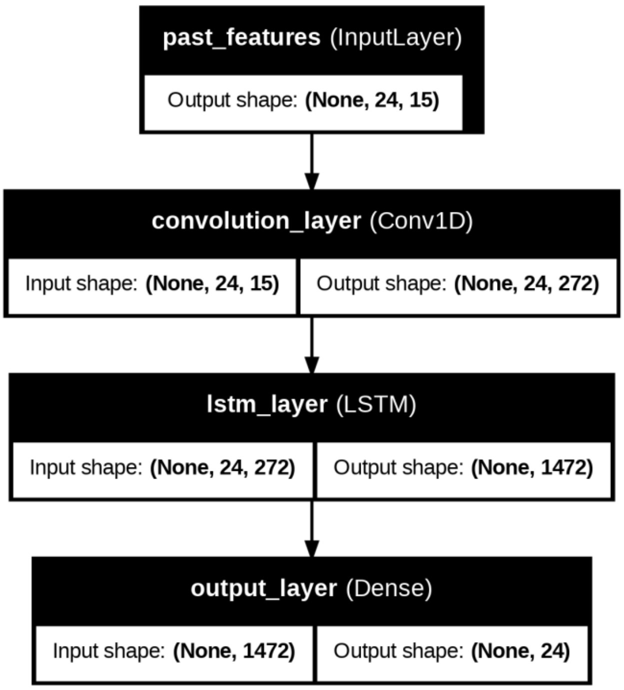


Fig. 6. Architecture of the neural network model using only historical data.

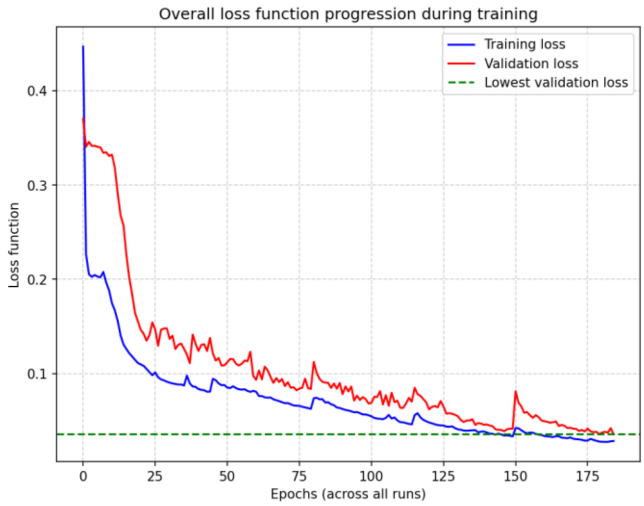


Fig. 7. Training and validation loss for the neural network model using only historical data.

Table 4. Error metrics for the Neural-based model using only historic data.

| Hour | nMAE [%] | MAE [kW] | MBE [kW] | $R^2$ [-] | MAPE [%] |
|---|---|---|---|---|---|
| 1. | 7.11 | 92.00 | 14.35 | 0.80 | 55.29 |
| 6. | 10.32 | 133.53 | 11.26 | 0.60 | 104.09 |
| 12. | 11.14 | 144.01 | 4.54 | 0.56 | 119.70 |
| 18. | 12.36 | 159.92 | 20.05 | 0.46 | 137.59 |
| 24. | 13.87 | 179.32 | 68.17 | 0.32 | 148.45 |

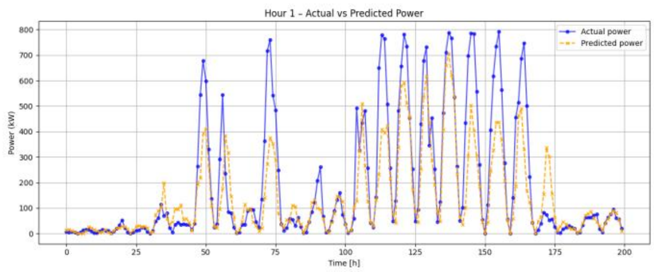


Fig. 8. Comparison of actual and predicted PV power for the first hour of the forecast horizon using the neural network model with historical data.

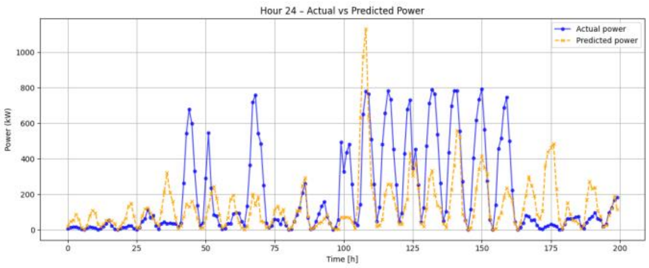


Fig. 9 Comparison of actual and predicted PV power for the last hour of the forecast horizon using the neural network model with historical data.

## 2.5 Model based on neural networks with the inclusion of weather forecasting

The final model incorporates both historical measurements and forecasted meteorological variables to improve short-term PV power predictions. A new branch was added to the model architecture to handle the predicted features, which expects a tensor of shape (24, 14) (see Fig. 10). This branch uses the same set of input features as the historical branch, with the exception of PV power, which is the target variable to be predicted. The validation loss curve again converges toward the training loss (Fig. 11), indicating that the model does not overfit.

The model achieves improved performance compared to the purely historical model, as summarized in Table 5. Prediction errors are lowest for the first hours of the forecast horizon (nMAE 6.05 %, MAE 78.25 kW, $R^2$ 0.86 for hour 1) and slightly increase toward the 24th hour (nMAE 6.88 %, MAE 88.99 kW, $R^2$ 0.82).

Fig. 12 and Fig. 13 show the comparison between the predicted and actual PV power for the first and last hours of the forecast horizon. The largest inaccuracies occur on days with rapidly changing cloud cover, when the model struggles to capture sudden power fluctuations. Conversely, on clear-sky days, the model predictions align closely with actual measurements. Representative examples of an overcast day and a clear-sky day are shown in Fig. 14 and Fig. 15, respectively.

A SHAP analysis was performed on the first 100 input sequences to assess feature contributions to the model predictions. The analysis was carried out for the first forecasted hour (Fig. 16) and the last forecasted hour (24th hour) of the prediction horizon (Fig. 17). Historical input sequences (24 preceding time steps) are shown in blue, while forecasted meteorological sequences (24 predicted time steps) are shown in orange. The results show that global solar radiation and previous PV power values have the strongest influence on the predictions. With increasing forecast horizon, the importance of historical PV power decreases, whereas the contribution of forecasted meteorological variables—particularly global solar radiation—becomes more dominant, reflecting the growing dependence of the model on weather forecasts.

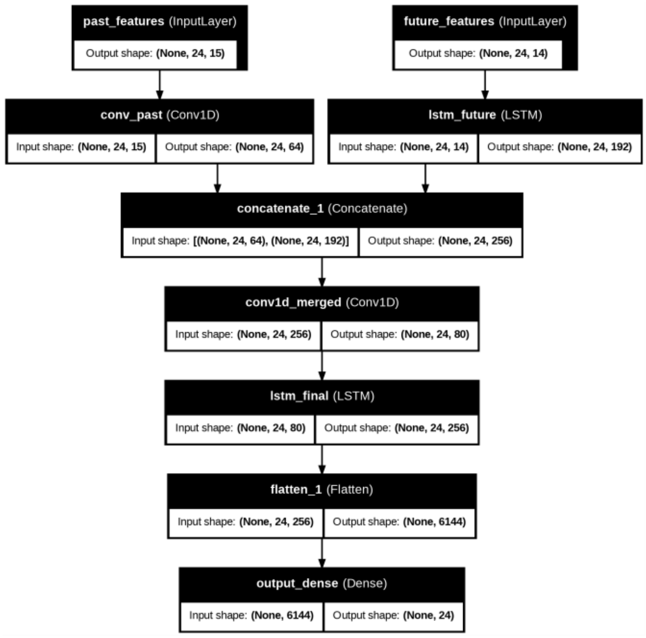


Fig. 10. Architecture of the neural network model using weather forecasting.

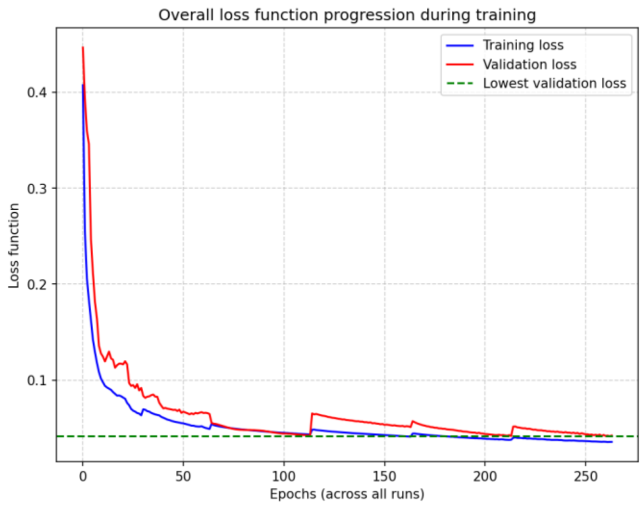


Fig. 11. Training and validation loss for the neural network model using weather forecasting.

Table 5. Error metrics for the Neural-based model using weather forecasting.

| Hour | nMAE [%] | MAE [kW] | MBE [kW] | $R^2$ [-] | MAPE [%] |
|---|---|---|---|---|---|
| 1. | 6.05 | 78.25 | -3.00 | 0.86 | 47.31 |
| 6. | 6.35 | 82.17 | 12.26 | 0.84 | 61.39 |
| 12. | 6.30 | 81.58 | -1.54 | 0.84 | 58.97 |
| 18. | 6.27 | 81.07 | 1.81 | 0.84 | 59.50 |
| 24. | 6.88 | 88.99 | -4.2 | 0.82 | 61.61 |

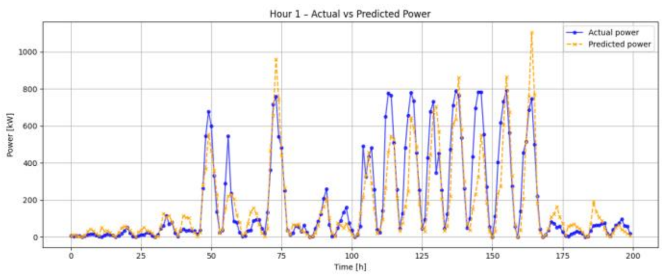


Fig. 12. Comparison of actual and predicted PV power for the first hour of the forecast horizon using the neural network model with weather forecasting.

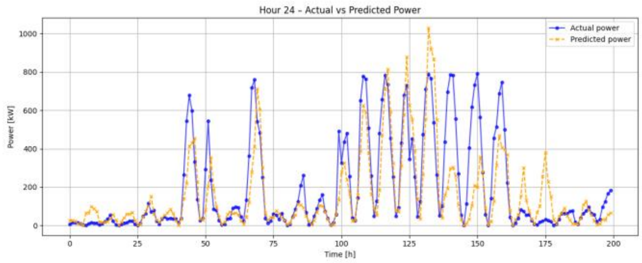


Fig. 13. Comparison of actual and predicted PV power for the last hour of the forecast horizon using the neural network model with weather forecasting.

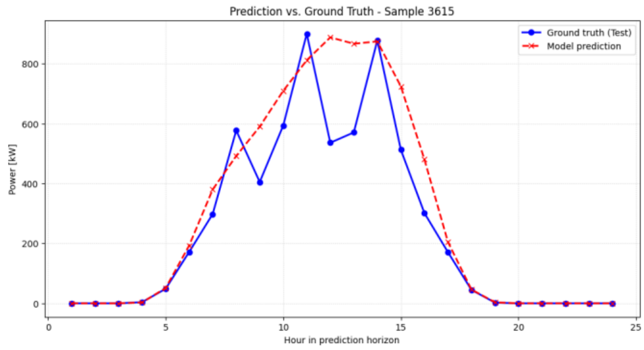


Fig. 14. Example of PV power prediction on a day with rapidly changing cloud cover, illustrating increased prediction errors caused by sudden irradiance fluctuations.

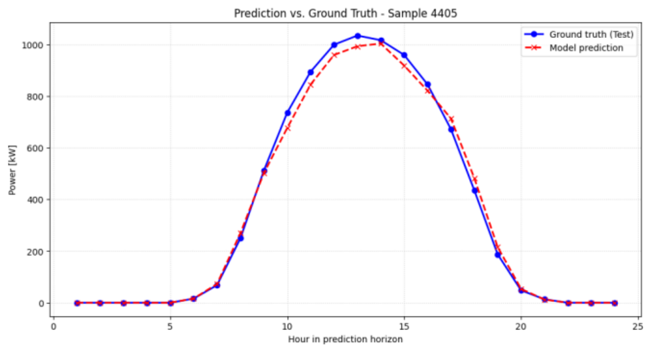


Fig. 15. Example of PV power prediction on a clear-sky day, showing close agreement between predicted and actual PV power output.

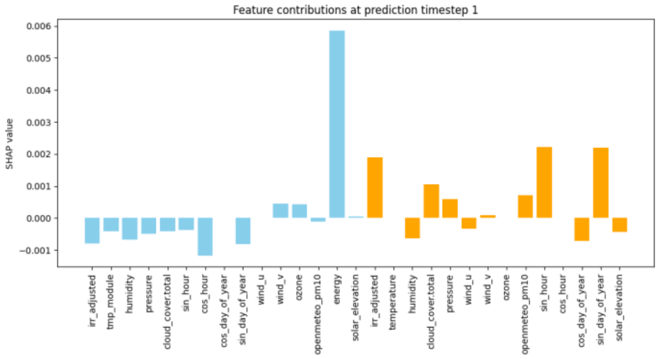


Fig. 16. SHAP feature importance for the first forecasted hour, showing contributions of historical (blue) and forecasted (orange) input features.

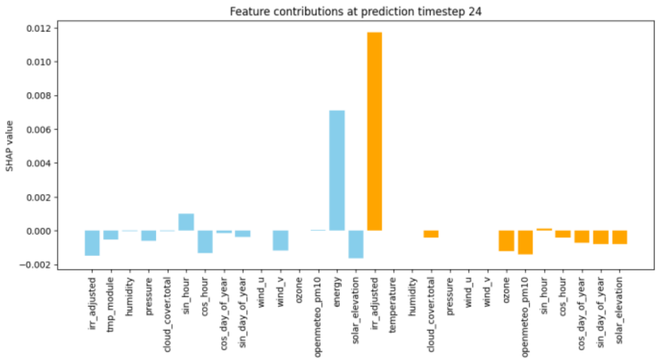


Fig. 17. SHAP feature importance for the 24th forecasted hour, highlighting the increasing influence of forecasted meteorological variables.

## 3 Problem Solution

The performance of all developed models was evaluated using standard error metrics, summarized in Tab. 1. As expected, Model 1 (historical hourly averages) achieves the poorest results, with nMAE = 12.48 %, MAE = 161.41 kW, MBE = −9.7 kW, $R^2$ = 0.51, and MAPE = 159 %. This high error reflects the limited predictive power of purely climatological averages, which do not account for current meteorological conditions.

The physical model based on Eq. (1) (Model 2) improves prediction accuracy compared to historical averages; nMAE decreases to 7.60 %, MAE is 98.35 kW, MBE is −59.7 kW, and $R^2$ = 0.80. The notable negative bias indicates systematic underestimation of the output power.

Further improvement is achieved by the physical model incorporating the incidence angle modifier (IAM) (Model 3), which accounts for the actual angle of solar radiation incidence on the PV panels. This modification reduces systematic bias and improves overall accuracy, yielding nMAE = 6.60 %, MAE = 85.43 kW, MBE = −10.5 kW, $R^2$ = 0.85, and MAPE = 82.32 %.

Neural network–based models provide substantially better results. Model 4, trained solely on historical data (24-h horizon), achieves an average nMAE of 11.26 %, MAE of 145.61 kW, MBE of 9.0 kW, $R^2$ of 0.54, and MAPE of 116.95 % over the entire forecast horizon. The results indicate that, without forecasted meteorological inputs, the model's accuracy is limited for longer horizons.

The most accurate predictions are obtained with Model 5, which combines historical measurements with forecasted meteorological variables (24-h horizon), achieves an average nMAE of 6.31 %, MAE of 81.70 kW, MBE of 3.13 kW, $R^2$ of 0.84, and MAPE of 58.79 % over the entire forecast horizon, indicating a closer agreement between predicted and measured values.

Overall, these results confirm that neural networks, particularly when combined with weather forecasts, are the most suitable approach for short-term photovoltaic power prediction, while traditional methods, such as simple historical averages or basic physical models, exhibit significantly lower predictive performance.

Table 6. Overview of evaluation metrics for all models.

| Model | Time horizon [h] | Avg. nMAE [%] | Avg. MAE [kW] | Avg. MBE [kW] | Avg. $R^2$ [-] | Avg. MAPE [%] |
|---|---|---|---|---|---|---|
| 1. | - | 12.48 | 161.41 | -9.7 | 0.51 | 159 |
| 2. | 1 | 7.60 | 98.35 | -59.7 | 0.80 | 77.30 |
| 3. | 1 | 6.60 | 85.43 | -10.5 | 0.85 | 82.32 |
| 4. | 24 | 11.26 | 145.61 | 9.0 | 0.54 | 116.95 |
| 5. | 24 | 6.31 | 81.70 | 3.13 | 0.84 | 58.79 |

## 4 Conclusion

In this work, a mathematical model for short-term photovoltaic power prediction was developed and tested. The model combines convolutional (Conv1D) and LSTM layers, with the highest accuracy achieved by integrating forecasted meteorological variables. This model reaches an average nMAE of 6.31 %, MAE of 81.70 kW, and coefficient of determination $R^2$ = 0.84, demonstrating close agreement between predicted and actual values. The model was implemented in TensorFlow, trained in the Google Colaboratory environment.

Various approaches were compared, from simple historical averages through physical models to advanced neural networks with different input combinations. The results show that classical methods, which do not account for current meteorological conditions, perform significantly worse than machine learning–based models.

Further improvements could be achieved by using shorter prediction intervals (e.g., 15 minutes), combining physical and data-driven models in an ensemble approach, and employing more accurate meteorological inputs. Locally calibrated forecast data could also enhance performance, especially on days with rapidly changing cloud cover.

The obtained results have direct practical applications in operational planning and optimization of PV power plant output. High-quality short-term predictions also have economic implications, including more accurate estimation of spot electricity prices and improved planning of energy supplies to the distribution network.

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## Contribution of individual authors to the creation of a scientific article (ghostwriting policy)

Jakub Hampejs carried out all aspects of the research, including conceptualization, methodology, data collection, data analysis, model development, experiments, and manuscript preparation.
Martin Libra provided valuable advice and guidance throughout the research process.
Václav Beránek provided access to the photovoltaic power plant data used in the study.